

**LabSync: An IoT-Based IT Laboratory Availability and Equipment
Monitoring System Using QR Codes at Bulacan State University –
Sarmiento Campus**

A Capstone Project by

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CHAPTER I

Today, universities are turning into smart campuses to make school operations faster, improve facility management, and provide better services for both students and instructors. One of the most important tools in this setup is the Internet of Things (IoT), where sensors and microcontrollers connect to the internet to collect real-time data and automate everyday tasks (Ye et al., 2024). Studies show that having connected devices gives school administrators accurate, up-to-date information on how rooms and equipment are being used throughout the day (Arunkumar et al., 2024). Along with IoT, Quick Response (QR) codes have become widely used in schools because they provide a quick, simple way for people to scan and access digital systems using their smartphones (Teves, 2024). Combining IoT hardware with easy QR code workflows helps academic institutions replace slow, traditional paper processes with a more responsive and organized setup.

Around the world, many universities already use smart monitoring setups to track classroom usage, power consumption, and equipment status. Research shows that these automated monitoring setups help departments manage facilities better by providing clear visibility into whether a room is open or occupied (Taruc&De La Cruz, 2024). At the same time, schools stick QR codes on equipment so users can report broken items directly, saving paper and cutting down reporting delays (Sağit et al., 2024). Using QR codes for reporting helps

technical staff find and fix issues faster, while keeping inventory records accurate and reducing the paperwork staff usually have to deal with (Sağıt et al., 2024).

Here in the Philippines, however, many colleges and universities still rely heavily on manual methods to handle laboratory schedules and computer equipment. Checking if a lab is available usually means walking over to read printed schedule sheets taped to the door, which are often out of date because of sudden room swaps or makeup classes. On top of that, professors often hand the physical lab key directly to the next teacher in the hallway without logging it at the department office, leaving administrators unsure of who actually has the key or if a room is being used. When computers break down, students have to write the problem in a paper logbook, and instructors or lab staff must manually type those notes into spreadsheets later. This manual setup takes a lot of time, risks losing important records, and often leads to multiple students reporting the exact same broken computer, which clutters the maintenance list and slows down repairs.

This exact situation happens at Bulacan State University – Sarmiento Campus (BuSU-SC), particularly in the IT Department. The computer laboratories across the IT Department building are used constantly every day for regular lectures, hands-on programming classes, and impromptu department activities. Even though schedules are created at the start of each semester, actual daily room use changes often, and physical keys are passed around without a clear log. At the same time, broken computer parts like defective

monitors, keyboards, or system units are still tracked by hand, adding extra work for faculty and slowing down the MIS staff when fixing equipment.

To solve these problems, this study developed **LabSync: An IoT-Based IT Laboratory Availability and Equipment Monitoring System Using QR Codes.**

LabSync uses an ESP32-powered box with electronic sensor slots that monitor whether each physical laboratory key is present or removed. When the key is inside the box and no class is ongoing, the web dashboard automatically displays the room as Available. Once an instructor takes the key, the system crossreferences their identity with the active timetable: if that instructor is assigned to that time slot, the room status switches to In Session; if an instructor pulls the key outside their assigned schedule, the status flags the room as Borrowed.

To handle hallway handoffs between classes, each lab key has an acrylic keychain tag with a QR code so an incoming professor can scan it with their phone to legally accept custody and claim the room. For broken hardware, each computer unit has a unique QR sticker that lets students report defective parts on their phones without needing an account, recording each submission as an individual ticket so the MIS staff can track specific workstation issues directly. Physical key custody is given to the user who is in the laboratory and who has access to the key (the authenticated user). This is the person who actually removed the key or received it via the Mobile QR code, rather than assuming it is being used by the scheduled faculty member. In addition, a server-side overdue

reminder is automatically sent directly to the actual current keyholder, 15 minutes after the end of the class.

Through web dashboards tailored for the IT Department Head, Program Coordinator, Faculty, and MIS Staff, the system provides real-time room availability, a collaborative Schedule Studio featuring working draft isolation and conflict detection, unified user management, and a centralized maintenance tracker. Official schedules remain protected under IT Department Head finalization authority while Program Coordinators assist in draft preparation. By bringing together IoT key tracking, mobile QR transfers, and verified digital fault reporting into one platform, LabSync removes paper logbooks, speeds up computer repairs, and makes lab management much more organized for the IT Department of Bulacan State University - Sarmiento Campus.

Background of the Study

For technical courses, the student's training facility is primarily the computer lab; therefore, ensuring the cleanliness of the computer lab and computer is vital for daily use. Many universities have endeavored to turn their campuses into "smart campuses" so as to realize the automation of room scheduling and equipment maintenance. Some schools overseas have more sophisticated systems such as surveillance cameras or facial recognition scanners, but these are too costly to deploy in Local state universities and are too intrusive for students and teachers to feel comfortable with respect to their privacy (Kunjiapu et al., 2025). This has led to the need for tools in schools that are easy to use, affordable, and can reliably track facilities without having to monitor people's movements (ElHaggar et al., 2023).

A computer lab in Philippine State Universities still requires some amount of manual papers. Logbooks, paper sign-up sheets, and paper and pen signatures are still used by most schools to track rooms and equipment usage. This configuration can easily result in errors, missed logins, and lost documents (Kunjiapu et al., 2025). When a student spots a broken computer, they'll typically jot it down on a piece of paper and then leave it there for days or weeks until it is entered into a spreadsheet. This delay delays repair, makes broken PCs sit in repair areas, and interferes with hands-on classes.

At Bulacan State University – Sarmiento Campus (BuSU-SC), particularly within the IT Department computer laboratories, faculty and students encounter significant friction regarding room visibility and hardware accountability. To verify whether a laboratory is vacant, instructors and students must walk across the building to check printed schedules posted on laboratory doors or look through windows. This manual process wastes valuable instructional time, especially when arranging makeup lectures, hands-on exam sessions, or departmental student consultations. Furthermore, physical keys are frequently handed over from one instructor to another in the hallway between consecutive class periods without returning to the department office. Without an immediate custody logging mechanism, administrators lose track of key possession and actual room occupancy. At the same time, broken monitors, keyboards, mice, or system units often sit unfixed for ages simply because manual paper reporting takes too long. To stop room lockouts and missing keys, LabSync lets users pass keys directly to one another using QR codes, backed by automated email reminders. The system

checks active daily schedules and adds a 15-minute grace period to the class end time (for instance, a 10:00 AM class gets a 10:15 AM return deadline). Once time runs out, an email goes straight to whoever actually holds the key—ensuring clear accountability and saving scheduled instructors from taking the blame for keys they don't have.

To address these challenges, the researchers developed **LabSync: An IoT Based IT Laboratory Availability and Equipment Monitoring System Using QR Codes**. The platform connects physical key access directly to a centralized web dashboard using an ESP32 key monitoring box equipped with electronic sensor sockets. Key dock status is continuously cross-referenced with academic timetables in real time: when a key is docked and no class is active, the room is marked as *Available*; when an authorized instructor pulls the key for a scheduled class, the room switches to *In Session*; when a key is retrieved outside scheduled hours, the room is classified as *Borrowed*. To solve the hallway handoff problem, each physical key is paired with an acrylic keychain featuring an embedded QR code, allowing the next instructor to scan the key with a smartphone and log custody immediately. For hardware maintenance, serialized QR code stickers are affixed to each desktop computer, enabling students to submit individual fault reports directly from their mobile devices without account registration, providing MIS technicians with a clear, direct log of distinct issues.

The LabSync project contributes to the United Nations Sustainable Development Goals (SDGs): SDG 4 (Quality Education), as it continues to keep

lab computers functional for students; SDG 9 (Industry, Innovation and Infrastructure), by implementing a practical IoT system for a state university; and SDG 12 (Responsible Consumption and Production), by replacing paper logbooks with a paperless, digital process.

In general, the manual system was very difficult for anyone to receive fast updates on the schedules of the rooms, key custody and broken computers at BulSU-SC. This absence of real-time information results in wasted trips around campus, confused schedules, and lengthy repair periods which negatively impact student learning.

This study is aimed at four major areas to correct the above problems:

1. Manual Availability Checks: Eliminate time wasted by students and teachers walking to the lab to check its availability.
2. Slow Repair Reporting: Replacing handwritten paper reports by allowing students to scan workstation QR codes to submit individual hardware fault tickets directly to the MIS office.
3. Scheduling conflicts in class - Avoiding scheduling double booked rooms during regular class, make up class or consultation sessions with a scheduling builder which will check for scheduling conflicts.
4. Key Tracking and Accountability: Automatically updates key ownership on the online dashboard, supports direct peer-to-peer QR code key transfers between back-to-back instructors, and sends automated email reminders whenever a key isn't returned within 15 minutes of class ending.

Objectives of the Study

To design, develop, and evaluate **LabSync: An IoT-Based IT Laboratory Availability and Equipment Monitoring System Using QR Codes** for Bulacan State University – Sarmiento Campus that integrates physical key tracking, mobile QR key custody transfers, individual workstation fault ticketing, MIS maintenance tracking, conflict-free academic scheduling, and enterprise security auditing.

The specific goals of the research are as follows:

1. Design and develop an IoT key monitoring box using an ESP32 microcontroller, 6.35mm jack sockets with resistor-divider sensing, an I2C 16x2 LCD screen, an optical GM65 barcode reader, and an active buzzer alarm to detect physical laboratory key presence and status in real time.
2. Develop a centralized web application that dynamically computes laboratory availability (Available, In Session, Borrowed), supports mobile QR key custody handoffs for faculty, executes automated server-side email return reminders after a 15-minute post-class grace period, provides zero-login computer fault reporting with individual ticketing for MIS staff, and incorporates role-differentiated access including MIS OJT intern management and a drag-and-drop schedule management studio with realtime visual collision detection.
3. Evaluate the system's performance and user acceptance using the

ISO/IEC 25010 software quality model and the Technology Acceptance Model (TAM) to ensure technical reliability and usability across the BuSU Sarmiento Campus community.

Significance of the Study

The development of **LabSync: An IoT-Based IT Laboratory Availability and Equipment Monitoring System Using QR Codes** provides an automated, reliable alternative to traditional paper logbooks and manual equipment tracking in state university computer laboratories. By combining IoT hardware key detection, live room availability computation, mobile QR key custody transfers, and individualized digital ticketing, this study establishes a practical, cost-effective framework tailored to the daily operational realities of an academic campus.

The findings and implementation of this study directly benefit the following stakeholders:

Students. The system removes the burden of manual paperwork by eliminating physical logbook sign-ins and paper equipment fault slips. Through workstation QR stickers, students can report broken hardware parts directly from their mobile phones without having to create an account or log in, ensuring their complaints are recorded immediately while allowing them to focus on hands-on laboratory activities using functional computers.

Faculty Members. Instructors can view real-time laboratory availability (Available, In Session, Borrowed) directly from their dashboards without walking to the building to check printed door postings. They can immediately monitor the condition and fault reports of PC units in their assigned laboratories online instead of relying on slow paper forms, allowing them to adjust hands-on seat plans ahead of class. Faculty members can also view and print their official teaching schedules prepared by the Department Head, and submit advance second-key reservation requests up to the following week's Saturday for authorized lab sessions. Furthermore, the mobile QR key transfer feature enables smooth, transparent hallway key handoffs between consecutive classes, eliminating trips to the department office while maintaining clear administrative accountability. Instructors also get automated email reminders if they're still holding a physical key 15 minutes after their class ends, reminding them to return or hand it off so the next class isn't locked out.

IT Department Head. The system streamlines timetable creation through an interactive Drag-and-Drop Schedule Studio with visual collision detection, while providing exclusive authority to finalize or reopen official master schedules and manage room editing locks. The Department Head oversees the unified User Management module to administer Faculty and MIS Staff accounts, including soft deactivation and replacement workflows enforcing one active technician. Additionally, the Department Head reviews and approves or rejects faculty second-key requests, tracks PC report follow-up inquiries on unresolved issues once per calendar day, assigns official curriculum courses into the grid, and generates printable class timetables complete with institutional signatory lines.

Program Coordinator. The system provides collaborative administrative access enabling the Program Coordinator to assist in developing semester schedule drafts, monitor instructor timetable assignments, review PC fault conditions, and manage faculty member accounts. Working draft isolation guarantees that changes made during coordination do not prematurely overwrite official class timetables until formally approved and finalized by the IT Department Head. The Program Coordinator can also request an additional laboratory key for authorized academic use, while remaining restricted from finalizing schedules, approving other key requests, or managing technical staff.

MIS Staff / Technical Custodians. Centralizing maintenance tickets into a single online queue eliminates delays caused by handwritten notes and manual spreadsheet encoding. The system registers individual fault tickets for each student report, allowing technicians to inspect detailed remarks, track unique diagnostic histories, repair defects faster, and monitor physical laboratory key custody. MIS staff can also set up and manage temporary accounts for OJT student interns with clear start and end dates. This lets interns handle day-to-day computer repairs while keeping admin controls, smart key boxes, and master schedules strictly off-limits. Account provisioning and replacement for MIS staff are managed exclusively by the IT Department Head to ensure administrative integrity. Plus, the system automatically logs every key transfer, giving staff a clear, verifiable audit trail without them ever having to hold the classroom keys themselves.

Bulacan State University – Sarmiento Campus. The implementation of LabSync accelerates the university's transition toward smart campus operations using cost-effective, maintainable open-source web technologies and microcontrollers. It optimizes facility utilization across all IT Department laboratories, enforces paperless workflows, and supports institutional sustainability goals.

Future Researchers. This study offers an empirical, verifiable blueprint for integrating ESP32 telemetry, web architecture, and QR code ecosystems within public educational institutions. It provides a reliable technical reference for developing non-intrusive asset monitoring and facility management solutions in resource-constrained environments.

Scope and Delimitations

Scope

This study focuses on the design, development, and implementation of LabSync: An IoT-Based IT Laboratory Availability and Equipment Monitoring System Using QR Codes, deployed comprehensively across all computer laboratories situated within the Information Technology Department building at Bulacan State University – Sarmiento Campus. These specialized instructional facilities are selected because they house the campus's computing infrastructure and host the highest concentration of hands-on technical coursework, rendering them particularly vulnerable to schedule overlaps, unmonitored room handoffs, and rapid peripheral wear.

The hardware architecture centers on a centralized key monitoring box powered by an ESP32 microcontroller, stationed directly within the department faculty office. The box utilizes 6.35mm jack sockets paired with resistor-divider analog sensing circuits to detect the presence or withdrawal of laboratory keys reliably. Physical operational feedback is delivered through an I2C 16x2 LCD screen, an active buzzer alarm that triggers if a key is placed into an incorrect slot, and an optical GM65 barcode reader. By cross-referencing key dock telemetry with active class timetables, the system dynamically calculates laboratory status into three operational states: *Available*, *In Session*, and *Borrowed*. To support rapid room transitions between consecutive classes, all physical laboratory keys are equipped with QR-coded acrylic keychains, allowing incoming instructors to

execute mobile key custody handoffs directly in the hallway without returning to the department office.

For equipment maintenance, every desktop workstation across all IT Department laboratories is labeled with an individual, serialized QR code sticker linked to a mobile-optimized web reporting interface. To strengthen reporting integrity while preserving friction-free access, students scan the workstation QR code followed by their Student ID QR code, automatically populating verified student identification without requiring account registration. Each submission is logged as an independent, individual ticket within the centralized maintenance module, enabling MIS technicians to review specific user remarks, inspect precise timestamps, trace recurring equipment fault patterns per computer unit, and update diagnostic statuses accurately.

The accompanying web portal provides dedicated, role-based modules: the IT Department Head utilizes an interactive drag-and-drop Schedule Studio equipped with visual collision detection and collaborative draft review, retaining sole authority to finalize or reopen official timetables, administer unified User Management (governing Faculty and MIS Staff replacement lifecycles), approve faculty second-key requests, and record once-per-calendar-day PC report follow-up inquiries; the Program Coordinator collaborates on draft timetables, monitors room availability, and manages faculty accounts; Faculty Members view live laboratory availability, monitor reported equipment defects in their assigned teaching rooms, execute peer-to-peer key handoffs, and submit advance second-key reservation requests; and MIS Staff track incoming individual repair tickets, supervise OJT intern accounts, and maintain an audit log of physical key custody. In the background, the server constantly checks active class schedules and system data to enforce a 15-minute key return window after each class, automatically emailing whoever is holding the key if it isn't returned on time.

Delimitations

This study is limited only to the computer laboratories inside the Information Technology Department building at Bulacan State University – Sarmiento Campus. Other areas, such as lecture rooms, science labs, and offices, are not included because the system is made specifically for computer labs that have many desktop units and shared class schedules.

For security reasons, the LabSync IoT Key Box does not save account details inside the device. Instead, it uses a Wi-Fi connection to check QR code scans directly with the main database in real time.

Because it needs an active connection to verify authorized users, the key box uses a **fail-secure** design. If the Wi-Fi connection drops, the device stops approving key requests, and trying to pull out a key while offline immediately triggers the hardware alarm. Once the Wi-Fi comes back on, the system automatically reconnects and syncs all current key statuses back to the database.

For the hardware, the system only monitors the physical keys using the key box and QR codes. It does not have automatic door locks, magnetic locks, or card-swipe door openers, meaning teachers still have to unlock the doors using the actual key. The project also does not use CCTV cameras or fingerprint scanners, keeping the system simple, low-cost, and non-intrusive.

For equipment reporting, the system only covers the physical parts of the desktop computers: monitors, keyboards, mice, and system units. It does not

include software repairs, reformatting, internet connection problems, or operating system troubleshooting.

Lastly, LabSync runs as a web-based system opened through a browser. The researchers did not create a downloadable mobile app for Android or iOS. Automated key return emails rely on campus internet and the school's mail server, meaning messages may be delayed if the network or mail server goes down. Additionally, OJT intern accounts are strictly limited to inspecting computers and updating ticket statuses (Pending, In Progress, Resolved). Interns cannot access the smart key box, transfer or claim keys, manage class schedules, or change user accounts and system settings.

Students can only use the system to report broken computer parts by scanning the QR code, and they do not need to create an account or log in to do so.

Definition of Terms

This section provides the operational definitions of technical, hardware, and domain-specific terms used throughout the study. Terms are arranged in alphabetical order based on their specific implementation within the LabSync system:

Active Buzzer. An auditory output module connected to GPIO25 of the ESP32 that sounds an alarm loop when a physical key is inserted into an incorrect dock slot.

Academic Term Engine: A calendar tool that automatically figures out the active school year (like 2026–2027) and semester (1st Semester, 2nd Semester, or Summer) based on the school's cutoff dates.

Automatic Key Return Reminder: A background service that automatically emails whoever has the lab key if it isn't returned within 15 minutes after class ends.

Collision Detection. An algorithmic check built into the Schedule Studio that identifies and visually flags time and room overlaps with red indicators before a subject block is dropped, preventing double-booked laboratories and faculty conflicts.

Current Key Holder: The database field that keeps track of the specific teacher or staff member who currently has the lab key.

Data Lag. The operational delay between the occurrence of a computer hardware failure and the formal logging and inspection of that issue by MIS staff under traditional, paper-based reporting workflows.

ESP32 DevKit V1. The central microcontroller board programmed in C++ (Arduino framework) that monitors physical key jack states, controls the LCD screen and buzzer, interfaces with the optical scanner, and transmits 5-second telemetry heartbeats to the web application over local Wi-Fi.

Grace Period: A 15-minute window after class ends before the system automatically sends a reminder email for an unreturned lab key.

GM65 Barcode/QR Reader. An optical scanning module interfaced with the ESP32 via UART serial communication to scan faculty authentication cards and room key QR codes directly at the key dock.

Internet of Things (IoT). A network of physical devices embedded with sensors, processing ability, and network connectivity that communicate and share telemetry with a centralized software application.

ISO/IEC 25010. An international standard for software quality evaluation used in this study to evaluate the system across functional suitability, performance efficiency, usability, and reliability.

OJT Account: A temporary account created by MIS staff with fixed start and end dates for student interns. It lets them update repair ticket statuses, but keeps smart key box controls and class schedules strictly off-limits.

6.35mm Jack Socket (Quarter-Inch Jack). The physical electromechanical port in the key dock that utilizes a resistor-divider circuit on the ESP32's analog-to-digital converter (ADC) pins to reliably detect the insertion or withdrawal of a specific room key.

Key Custody Transfer. The mobile process wherein an incoming instructor scans the QR tag on a physical key keychain to digitally assume responsibility for the key and room during class transitions without visiting the department office.

Perceived Ease of Use (PEOU). A metric within the Technology Acceptance Model (TAM) assessing the degree to which students, faculty, and administrative staff find the LabSync web portal and reporting interfaces effortless to navigate.

Perceived Usefulness (PU). A metric within the Technology Acceptance Model (TAM) assessing how much users believe LabSync enhances laboratory availability visibility, scheduling efficiency, and maintenance turnarounds.

Quick Response (QR) Code. A machine-readable optical label utilized in two ways within the system: serialized stickers on desktop computers for zero-login student fault reporting, and two-sided acrylic keychain tags for mobile key handoffs and key dock identification.

Room State. The dynamic operational condition of a laboratory determined by cross-referencing physical key presence with active schedule timetables, classified into three discrete values: *Available*, *In Session*, or *Borrowed*.

Schedule Studio. The interactive drag-and-drop timetable module utilized by the Collaborative Master Schedule / Draft Schedule. A distinct working timetable state stored in schedule_drafts allowing concurrent, room-locked schedule preparation by the IT Department Head and Program Coordinator without affecting published official class schedules until explicit finalization.

PC Report Follow-Up. An administrative tracking mechanism enabling the IT Department Head to record structured follow-up inquiries on unresolved maintenance tickets once per calendar day, logging timestamp and user attribution in the maintenance records.

Program Coordinator. An academic administrative user role responsible for collaborating with the IT Department Head on draft laboratory schedule creation, managing faculty profiles, and requesting additional keys, operating with restricted administrative privileges that prevent finalizing official schedules or managing MIS personnel.

Student ID QR Verification. A dual-stage optical verification workflow wherein a student scans a laboratory workstation QR code followed by their institutional Student ID barcode/QR code, automatically extracting authenticated student identification and generating a temporary, replay-protected verification session without requiring account registration.

User Management. A centralized administrative module that consolidates account lifecycle operations for Faculty and MIS Staff under the IT Department Head, and OJT Interns under MIS Staff.

CHAPTER II

REVIEW OF RELATED LITERATURE AND SYSTEMS

The aim of this chapter is to provide an overview of previous studies, conceptual frameworks and existing operational platforms that are directly related to the development of LabSync: An IoT Based IT Laboratory Availability and Equipment Monitoring System Using QR Codes. It provides the theoretical background and technical means for physical microcontroller telemetry, mobile QR code interactions and web-based scheduling to be integrated. In addition, the chapter examines the various enterprise and commercial systems in place and pinpoints their operational weaknesses that warrant the development of a solution that is localized, cost effective and relevant to the context of a Philippine state university.

Academic institutions all over the globe have greatly improved operational efficiency by leveraging Internet of Things (IoT) technologies. Campuses can now collect, process and analyze data on the facilities in real time by connecting microcontrollers, hardware sensors and cloud systems. According to Ye et al. (2024), the smart campus architecture based on IoT technology helps to optimize the management of the campus. Using a unified dashboard, people can monitor the status of classrooms and the condition of the equipment. Arunkumar et al. (2024) successfully showed how student and campus information systems using the IoT reduce the amount of paperwork in education and create real-time institutional intelligence.

Centralized telemetry provides education leaders with evidence-based data to guide decisions, efficient use of instructional spaces, and reduced waste of resources. The researchers Mallari et al. (2025) in the Philippines were able to verify that the use of IoT and automated tools in smart classrooms will increase the utilization of educational resources, automate repetitive monitoring tasks, and reduce any time delays required during instruction resulting from administrative delays. In a similar fashion, Taruc and De La Cruz (2024) studied the use cases of the IoT in the educational system in the Philippines and concluded that the use of open-source micro-controller and the standard network telemetry will provide viable solution for local educational campuses in terms of scalability, cost-effectiveness, and long-term solutions. Their results highlight that automated tracking of facilities is possible within state budgets and can be achieved through the design of systems that account for local budgets.

The conventional method of tracking and maintaining the system in the laboratory is based on manual paper logbooks and physical inspection sheets, which create delays in data recording, errors, and delayed repairs. Jadhav et al. (2024) pointed out that manual inventory logging can hinder the productivity of the operations, and suggested the use of a QR-based asset inventory management platform where each asset within a facility is given its own optical tag. This enables staff to use a handheld mobile device to scan the codes, resulting in real-time updates of the database and helping to reduce duplicated paper forms, while providing a more accurate tracking system. This was complemented by the work of Rabiah et al. (2022) who identified the bottlenecks

of manual equipment borrowing and hardware checking, and found that a web-based system with QR code scanners could ease the custody transfer process while maintaining good audit trails.

Sağit et al. (2024) confirmed that QR code mechanisms are widely used in learning environments with high usability, as they have no specific requirements for hardware usage and can be used by students and teachers without expertise. With the integration of QR code interfaces and automated notification modules, Teves (2024) reiterated how this combination significantly cuts down maintenance response times, preventing equipment failures from impacting instructional delivery.

Although there are many operational benefits to the IoT applications, design is paramount and balancing institutional oversight with user privacy and simplicity of operation is a critical design challenge. El-Haggag et al. (2023) found that the use of non-invasive tracking methods, rather than pervasive surveillance systems like facial recognition cameras or continuous biometric scans, can lead to greater uptake of campus IoT platforms. Non-intrusive sensory approaches are sufficient to meet administrative monitoring needs, but reduce privacy issues and ethical dilemmas in the academic environment.

Kunjiapu et al. (2025) observed that a key problem with digital transformation in Philippine public education is the intricacy of user interfaces and authentication processes. Adoption barriers are eliminated and high

reporting compliance is achieved with systems that support frictionless interaction, such as by using public QR scan endpoints without requiring students to create accounts.

Taking into account these views, the literature validates that an IoT framework, which integrates a discrete physical key sensing capability, low friction mobile QR codes interfaces and localized web dashboards, is an effective and affordable solution to the operational challenges of computer laboratories.

Review of Related Systems

Cisco. (Cisco Systems, Inc., n.d.) Cisco Systems, Inc. created the Cisco Smart Campus solution. Cisco Smart Campus is a campus management system with IoT (Internet of Things) features that enhances the operational efficiency, safety and facility monitoring within educational institutions. The system includes connected sensors, network infrastructure and cloud-based analytics that record and track the occupancy of learning spaces, energy usage, environmental conditions and security systems in real time. It allows administrators to have a single view of the building usage, optimize resources and improve campus-wide connectivity. Cisco Smart Campus also has features such as wireless networking, smart lighting, surveillance, and automated alerts, which facilitate data-driven decision-making and smart facility management.

The system offers the advanced analytics and real-time monitoring capabilities needed for large scale campus operations. It has a scalable

enterprise architecture to allow for integration between buildings and departments. However, it is costly, and complex infrastructure and technical expertise are necessary for implementation, making it more difficult for smaller institutions. While Cisco Smart Campus was offered to monitor labs remotely with cameras and higher costs, LabSync is offered for low cost, localized laboratory monitoring via a key monitoring box and QR based reporting which makes it more practical for resource constrained universities.

IBM. (IBM Corporation, n.d.) IBM Intelligent Campus Management system is a product of IBM Corporation. IBM Intelligent Campus Management is a smart campus solution that enhances operation efficiency and facility management in higher education institutions. The system offers predictive maintenance and asset-tracking capabilities that enable administrators to see in real time if equipment is performing well or if there are any issues with assets in the campus. It leverages Internet of Things (IoT), Artificial Intelligence (AI) and cloud computing technologies to analyze data, minimize equipment downtime, and optimize resource use. IBM Intelligent Campus Management provides predictive analysis that enables institutions to avoid equipment failures before they occur, and can integrate enterprise data from various campus systems.

However, it is not suitable for smaller institutions with limited budgets due to its high deployment cost, complex infrastructure and enterprise architecture. LabSync, on the other hand, is a real-time lab availability monitoring and zero login QR computer fault reporting solution, which is cost-effective and practical for State Universities with limited resources.

Google. (Google LLC, n.d.) QR-Integrated Google Forms is a web-based digital reporting application developed by Google that allows users to create their own online forms via QR code. The system enables schools and organizations to collect the data in an electronic way, where QR codes are linked to reporting forms, thus making data collection easy and without any paper. It is 100% online and receives submissions in real time, directly into cloud-based spreadsheets. Its simplicity, wide access and low cost are its greatest assets. Easy to use, small technical installation and free to use, it is widely used in low-budget institutions.

However, the platform is missing the integration capability with IoT devices, real-time room occupancy detection, and automated maintenance workflows. It is a static digital form that is not hardware-based, and has no diagnostic tracking features. LabSync, on the other hand, is an IoT solution that integrates physical key presence detection with a web-based dashboard for real-time computer equipment condition monitoring, providing a more comprehensive and automated laboratory management solution.

Skedda. (Skedda, n.d.) Skedda Room Booking System is a cloud-based scheduling solution designed by Skedda for universities, offices and organisations to use online to book rooms. The system provides instant booking via an easy-to-use interface, users can check availability and book it easily via web access.

The platform is highly functional in visualizing booking in rooms and gives a user-friendly experience with little training for the user. Skedda's emphasis, however, is solely on reservations. Does not track equipment use or utilize IoT

sensors to verify if a room is actually being used. This restriction does not allow it to check if the actual room is occupied or if there are any hardware problems within the premises. In contrast to the simple scheduling view, LabSync can also set up for key detection to verify lab occupancy, and students can use QR codes to report on the current equipment status, filling in the gaps in monitoring room use and lab equipment.

Brightly Software. (Brightly Software, Inc., n.d.) Brightly Maintenance (formerly SchoolDude) is a cloud-based computerised maintenance management system (CMMS) for school management of maintenance requests and tracking physical assets. It lets administrators and staff log work orders, track maintenance processes, and keep track of equipment and facility conditions.

One of the key benefits of the platform is its structured maintenance workflow management and asset tracking features, enabling institutions to effectively manage repair requests and track equipment history. The platform does not however have zero-login, QR-based equipment-level reporting nor does it support monitoring of IoT keys for real-time detection of facility use. Unlike LabSync, which integrates IoT hardware for detecting lab availability and reports on the condition of computers with a QR code, Brightly Software is more of a work order tracking and asset log software.

LabWare. (LabWare, Inc., n.d.) The LabWare Laboratory Information Management System (LIMS) is an enterprise automation platform to manage laboratory data, sample workflows and inventory processes. The system can be integrated with the bar code and QR code for laboratory equipment and sample

tracking. It offers centralized reporting, automated documentation, and secure database management, with a focus on research labs, universities, and industrial testing facilities. The system's robust data management and regulatory compliance capabilities are among its key advantages and can help streamline complex laboratory workflows.

The deployment and maintenance of the system is however costly and is not feasible for smaller departmental setups. Its high level of clinical features necessitates expertise in technical training for administrators to use it correctly. Furthermore, it is designed specifically for research and sample testing in the laboratory, not for instructional use in a computer laboratory. Unlike LabWare LIMS, LabSync is a narrow-spectrum product designed for use in university IT laboratories, with its key presence detection using ESP32 for occupancy tracking, and QR codes for computer hardware maintenance reporting.

Comparison of Related Systems

Table 1. Comparison of Related Systems

Features	Cisco	IBM	Google	Skedda	SchoolDude	LabWare	LabSync
Real-Time Room Occupancy Monitoring	✓	✓					✓
IoT-Based Hardware Integration	✓	✓					✓
Smart Key-Based Detection							✓
QR Code Equipment Reporting							✓
PC Condition Monitoring		✓			✓	✓	✓
Room Claim		✓					✓
Schedule Management Integration	✓			✓	Partial	Partial	✓
Cost-Effective for State Universities			✓	✓	Partial		✓

Based on the comparison table, there is a clear gap between expensive enterprise systems like Cisco and IBM, and simple digital tools like Google Forms and Skedda. While enterprise systems have powerful features, they cost too much for state universities to buy and maintain. On the other hand, tools like Google Forms and Skedda are affordable, but they do not have IoT hardware integration to detect physical room keys or check real-time occupancy.

LabSync fills this gap by bringing together IoT hardware and affordable web features in one system. Designed for the computer laboratories in the IT Department building, it is the only system among the choices that combines smart key-based detection, QR code equipment reporting, PC condition monitoring, room claiming, and conflict-free schedule management. LabSync offers a practical and low-cost solution that gives the university reliable hardware monitoring and administrative tools without the heavy cost of enterprise software or the delays of manual logbooks.

Synthesis

Based on the literature review and system analysis, it was revealed that QR code and Internet of Things (IoT) technology are the effective tool for modernizing the school/laboratory management. Research studies highlight the benefits of automated tracking and monitoring web dashboards to increase daily efficiency by providing real-time information about room and equipment condition.

But in between the available options there is clearly some gap as seen in the comparison table. Large enterprise systems such as Cisco and IBM are too costly, complicated and difficult to support for local public schools. However, there are some simple tools, such as Google Forms and Skedda that are free or inexpensive, but may not include IoT hardware integration to detect a physical room key or confirm actual lab use. This indicates that the platforms available are too expensive, or do not provide the hardware capabilities required of a university computer laboratory. For schools that have limited budgets, a monitoring system should be cost-effective, user-friendly, and should be able to connect physical key movement with online records. Manual logbooks introduce data-lag, impacting on computer repair time, double rooms and key management.

LabSync delivers this solution with affordable IoT hardware and an online management system. It features an ESP32 key monitoring box for detecting room keys, an interactive drag-and-drop scheduling tool for visual conflict notifications, and mobile QR key handoffs for faculty. Students can scan QR stickers on desktop computers without logging in to send individual computer fault reports to technical staff, providing a clear and direct list of issues for technical staff to resolve. LabSync's QR code reporting and physical key discovery makes for an easy-to-use, cost-effective solution that helps state universities to efficiently manage their computer labs without the investment in expensive enterprise software. While commercial tools handle basic online bookings or standard helpdesk tickets, LabSync bridges the gap between what

happens on screen and inside the physical lab. By linking real-time physical key detection, automated class schedules, and post-class custody reminders, LabSync makes sure computer labs in state universities are properly handed over and kept secure after every single class.

CHAPTER III

DESIGN AND METHODOLOGY

This chapter presents the design and development of LabSync. It covers the conceptual framework, development methodology, functional and non-functional requirements, system architecture, database design, and IoT hardware integration. It also discusses the system workflow, implementation procedures, and testing and evaluation methods used to assess the system's functionality, reliability, security, usability, and user acceptability.

Requirement Analysis and Documentation

In this section, presents the functions and quality standards required for LabSync system to effectively address the manual struggles of laboratory management of Bulacan State University – Sarmiento Campus.

Table 2. Functional Requirements

ID	Functional Requirements	Description
FR-01	User Authentication and Role Management	
FR-02	Laboratory Room Monitoring	The system shall display the current availability and status of laboratory rooms.
FR-03	Laboratory Schedule Management	The system shall allow the IT Department Head and Program Coordinator to collaboratively create and edit schedule drafts with conflict detection and room lease locks, while restricting official schedule finalization and reopening exclusively to the IT Department Head.

FR-04	PC Fault Reporting and Maintenance	
FR-05	Physical Key Management	The system shall allow MIS staff to register, monitor, and update the status of laboratory keys.
FR-06	Key Transfer and Room Claim	
FR-07	IoT Laboratory Monitoring	The system shall receive data from the ESP32 device to monitor key presence, room activity, and device connectivity.
FR-8	System Records and Audit Logging	The system shall record important activities and maintain records of user actions, key transactions, maintenance activities, and system events.
FR-09	Automated Key Return Reminder	The system shall check active class schedules every minute and automatically send an email reminder to the current key holder when the laboratory key remains unreturned 15 minutes after the scheduled class end time, while preventing duplicate reminders.
FR-10	OJT Account Management	The system shall allow MIS Staff to create and manage temporary OJT intern accounts with defined internship dates and permit authorized interns to inspect and update repair tickets while restricting access to the IoT key box and Schedule Studio.

The functional requirements table outlines the core features and daily operations that the LabSync system is designed to perform. As detailed above, these include managing multi-role user access, collaborative schedule drafting with official finalization controls, monitoring laboratory schedules, tracking physical keys and advance second-key requests, and integrating with IoT hardware. These requirements ensure that administrative heads, academic coordinators, faculty members, and MIS technicians can seamlessly handle routine operations - such as dual-stage QR hardware reporting, one-active-technician account lifecycle management, and automated return reminders - while the system rigorously logs all laboratory transactions.

PC faults via QR codes or receiving automated key reminders—while the system securely logs all laboratory activities.

Table 3. Non-Functional Requirements

ID	Non-Functional Requirements	Description
NFR-01	Performance	The system shall provide quick responses and timely updates when users access laboratory information and perform system functions.
NFR-02	Usability	The system shall provide a simple and organized interface that allows faculty, MIS staff, and administrators to use the system with ease.
NFR-03	Security	The system shall protect user accounts and system data through crypt password hashing using 12 salt rounds, session inactivity timeouts, role-based access control (RBAC), parameterized SQL queries to mitigate SQL injection risks, and centralized security audit logging.
NFR-04	Reliability	The system shall maintain accurate records and perform system operations consistently without unnecessary errors or data conflicts by using atomic database claims for reminder deduplication and a retry mechanism with a 3-minute retry interval for network-dependent email transmission.
NFR-05	Compatibility	The system shall be accessible through commonly used web browsers and mobile devices for supported laboratory activities.
NFR-06	Scalability	The system shall be designed to accommodate additional users, laboratory rooms, schedules, and IoT devices as the laboratory's needs increase.

The non-functional requirements define the technical standards and quality attributes the LabSync system must maintain during operation. As detailed below, these encompass six key areas: performance, usability, security, reliability, compatibility, and scalability. Together, these criteria ensure the platform is highly responsive, secure, accessible across supported devices, and structurally capable of accommodating future growth in users, schedules, laboratory rooms, and IoT hardware.

Table 4. Hardware Requirements Table

Component	Specification	Purpose
ESP32 Dev Module	ESP32 development board with built-in Wi-Fi and programmable GPIO pins	It processes input from the QR scanner and key detection sockets, controls the LCD and buzzer, and communicates with the system server through Wi-Fi.
GM65 QR Code Scanner	Serial communication interface; 9600 baud rate;	Scans and reads QR codes used for user verification and laboratory access.
	connected to GPIO 17 (RX) and GPIO 16 (TX)	The scanned data is transmitted to the ESP32 for processing.

I2C LCD 16×2	16 columns × 2 rows; I2C interface; GPIO 21 (SDA) and GPIO 22 (SCL); I2C address commonly 0x27	Provides visual feedback to users by displaying system status, instructions, verification messages, and access results.
6.35 mm Jack Socket	6.35 mm (¼-inch) jack socket; configured using INPUT_PULLUP; connected to GPIO 32 for Room 203 and GPIO 33 for Room 204	Functions as the key-detection mechanism. It allows the ESP32 to determine whether a laboratory key is inserted or removed from the designated key slot.

The hardware requirements of the LabSync system consist of the components needed to support the monitoring and access functions of the laboratory. The ESP32 Dev Module serves as the main controller, processing the information received from the QR code scanner and key detection sockets while also managing the LCD and buzzer. The GM65 QR Code Scanner is used to read and verify the QR codes of users before granting laboratory access. The I2C LCD 16×2 provides immediate visual feedback by displaying instructions, system status, and verification results. The 6.35 mm Jack Sockets are used for detecting whether the laboratory keys are inserted or removed from their designated slots.

Lastly, the active buzzer provides audible notifications for important system events, such as successful verification, key detection, and errors. Together, these

hardware components allow the LabSync system to perform its intended laboratory monitoring and access-control functions in an organized and efficient manner.

Conceptual Framework

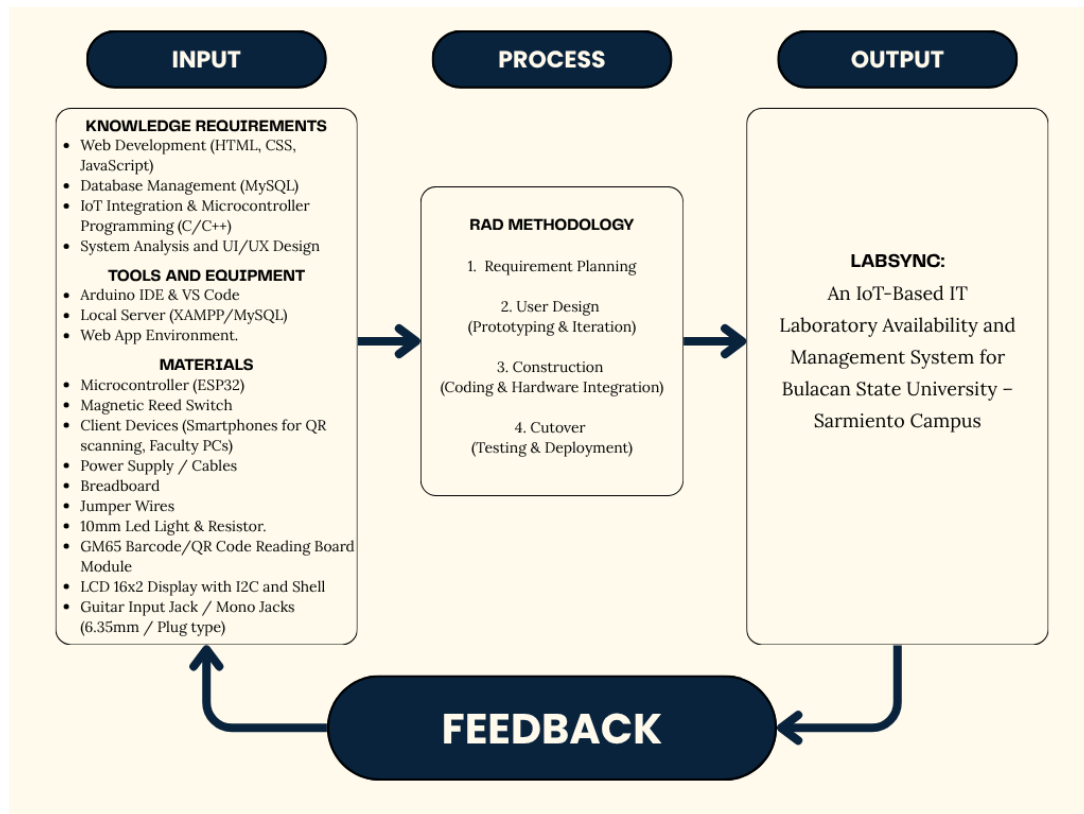


Figure 1. Conceptual Framework

This study has a conceptual framework that is based on the Input-Process-Output-Feedback (IPOF) model. It is a model that demonstrates the movement of the resources, the use of the development methodology, and the resulting system that will be used in Bulacan State University – Sarmiento Campus.

Input

The Input phase involves the preconditions required to start the development of LabSync.

Knowledge Requirements: This includes the technical knowledge needed for the development of the system, such as web development using HTML5, CSS3, JavaScript (ES6+), and Node.js with Express.js, relational database management using MySQL/MariaDB, IoT integration and microcontroller programming using C/C++ (Arduino framework), and system analysis and UI/UX design.

Tools and Equipment: The software and development environment consist of the Arduino IDE for programming the hardware, VS Code for developing the web application, the Node.js runtime environment with npm package management, the MySQL/MariaDB relational database engine, and web browsers for testing.

Materials: These include the physical components needed to build the IoT key dock, such as the ESP32 Dev Module microcontroller, 6.35mm jack sockets paired with resistor-divider sensing circuits, client devices (smartphones and PCs), power supply and cables, optical GM65 Barcode/QR Code Reading Module, I2C 16x2 LCD display, active piezo buzzer, and custom acrylic key enclosure. LED lights and resistors, GM65 Barcode/QR Code Reading Board Module, LCD 16x2 Display with I2C and Shell, and guitar input jack or mono jacks

Process: The Process stage incorporates the Rapid Application Development (RAD) model. It was selected because the method allows the researchers to develop, test, and improve the system through different stages.

Requirement Planning: Determining the scope of the system and identifying the requirements needed for the laboratory management system.

User Design: Creating the system prototypes, UI mockups, and hardware design based on the identified requirements.

Construction: Developing the Node.js and Express.js backend RESTful API, MySQL/MariaDB relational database schema, responsive web application interface, and ESP32 firmware in C++ using the Arduino framework, followed by physical hardware integration.

Cutover: Conducting the final testing, deployment, and implementation of the system in the laboratory.

Output

The final output is the completed system “LabSync: An IoT-Based IT Laboratory Availability and Management System for Bulacan State University – Sarmiento Campus.” The system is designed to provide real-time monitoring of laboratory availability, equipment reporting, and centralized laboratory management through the use of web-based and IoT technologies.

Feedback

The Feedback loop provides a way for the system to be continuously improved. The results of testing, together with the observations and recommendations from the intended users, can be used to identify parts of the system that need improvement. This allows the researchers to make necessary changes to the system requirements, hardware configuration, and other features to better suit the needs of the department.

Development Methodology

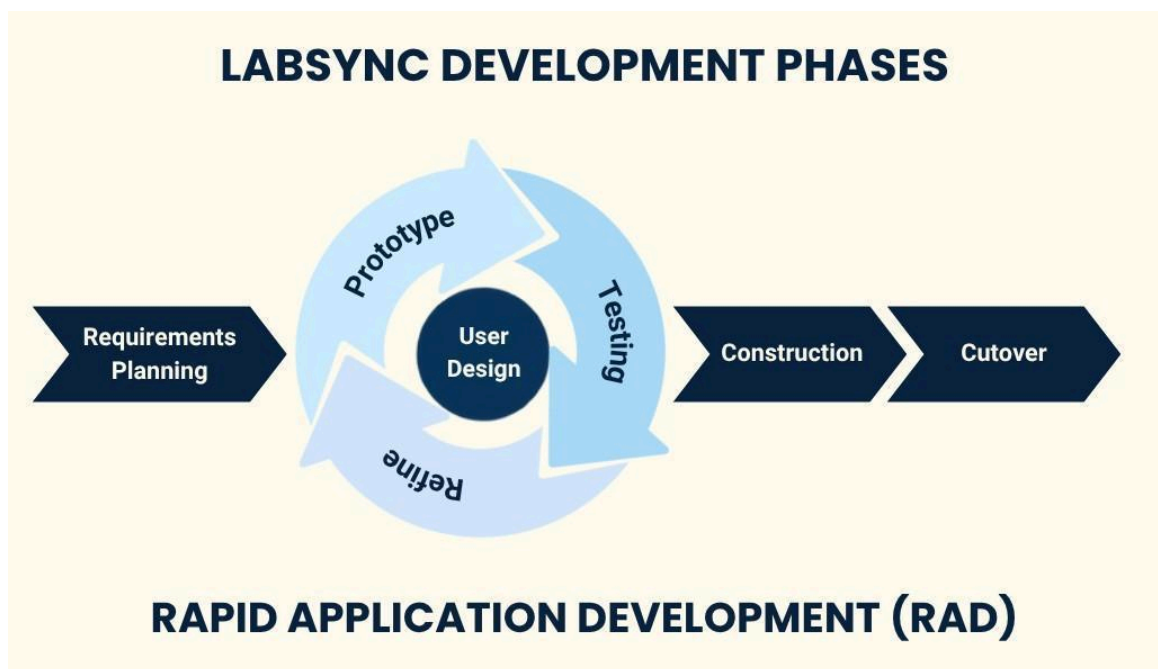


Figure 2. Development Methodology (RAD)

The researchers used the Rapid Application Development (RAD) method to build LabSync. This approach was chosen because it allows the team to

quickly create working prototypes, test them with users, and make changes based on their feedback. As shown in the diagram, the development process has four main steps: Requirements Planning, User Design, Construction, and Cutover.

In **Requirements Planning**, the researchers looked into the daily problems inside the IT computer labs, such as missing key records, room schedule conflicts, and slow PC repair reporting. From there, the team listed the exact features, hardware parts, and software tools needed for the project.

The **User Design** stage was done in repeated cycles of prototyping, testing, and refining. The team created early layouts of the website and basic test circuits for the ESP32 key box. By showing these mockups to teachers and staff, the researchers saw what worked and what was confusing, allowing them to adjust features like the drag-and-drop schedule and the mobile QR scanner before building the final version.

In the **Construction** phase, the actual system was built. The web application and RESTful API were developed using Node.js, Express.js, JavaScript, and MySQL/MariaDB, while the ESP32 firmware was written in C++ using the Arduino framework. At the same time, the physical acrylic box was assembled with its 6.35mm key jacks, LCD screen, buzzer, and QR scanner.

In the final phase, the **Cutover** stage involved bringing the finished system to the IT Department for testing and evaluation. Students, faculty, and staff tested the website and the key box. Their feedback was measured using the ISO/IEC

25010 software standard and the Technology Acceptance Model (TAM) to see if LabSync was reliable, easy to use, and ready for actual campus use.

System Design

The system design serves as the overall blueprint showing how the software and hardware components of LabSync connect and operate together. Rather than functioning solely as a typical website, the system links an online dashboard directly with an electronic key monitoring box so that room availability and key custody update immediately. This setup directly replaces manual logbooks and door schedules, establishing a more accurate and organized process for tracking laboratory rooms and computer equipment.

To clearly illustrate how the system is structured, several design models are presented. The Context Diagram outlines the general system boundary and shows how the platform interacts with its external entities: the IT Department Head, Faculty, Students, MIS Staff, and the ESP32 key box. The Use Case Diagram defines the specific actions and access permissions granted to each user role. The Data Flow Diagrams (DFD) trace the movement of information across the system, mapping how schedule updates, key presence signals, and workstation fault reports travel from user interactions into the database.

For the data and hardware structure, the Entity-Relationship Diagram (ERD) defines the database tables and relationships used to store user accounts, class timetables, workstation assets, key logs, and repair tickets. The System

Architecture Diagram illustrates the hardware-to-software pipeline, showing how the ESP32 microcontroller communicates sensor events to the web server over local Wi-Fi. Detailed Flowcharts document the step-by-step logic of essential operations, including dynamic room status calculation, mobile QR key custody transfers, visual schedule conflict detection, and individual fault reporting. Together, these diagrams provide a complete overview of how LabSync connects physical laboratory tracking with centralized web management.

Context Diagram

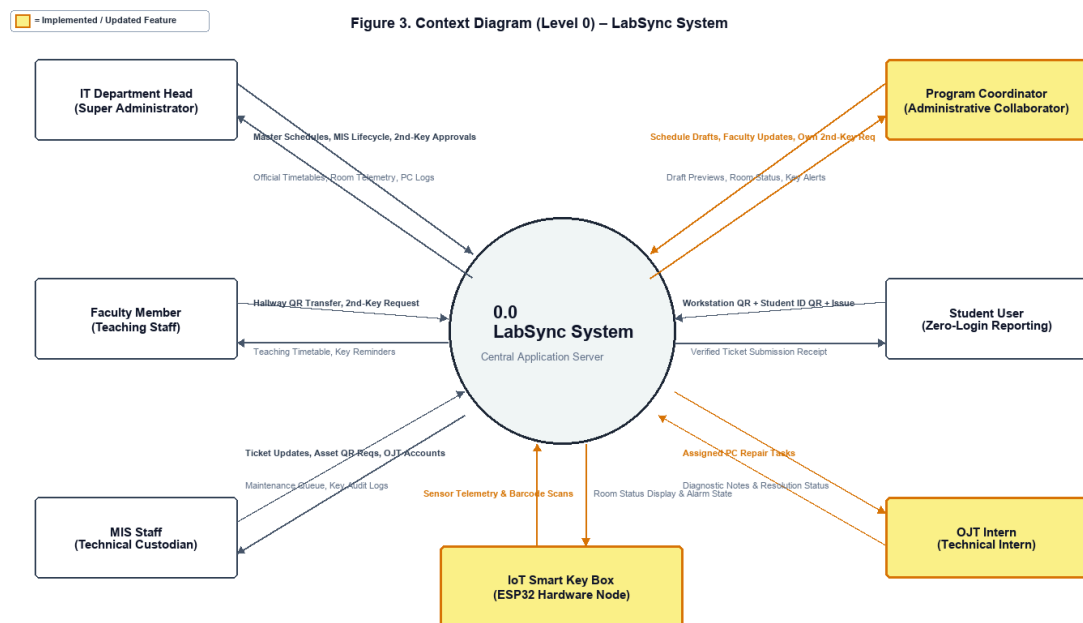


Figure 3. Context Diagram

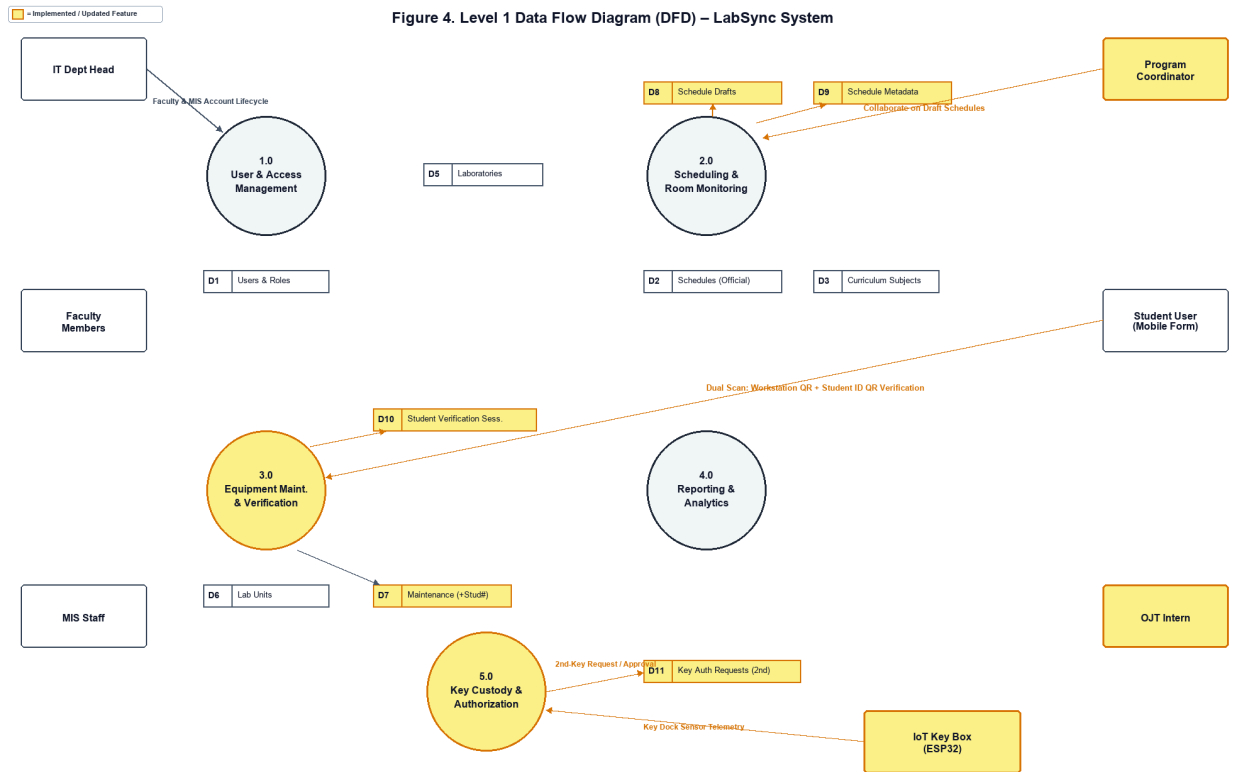
The Context Diagram provides a high-level overview of the boundary and data interactions of the LabSync system with its four external entities: the Department Head, Faculty, MIS, and Student. MIS staff also manage OJT interns

Figure 3 illustrates the Context Diagram (Level 0) of the LabSync system, detailing the interactions between the centralized software platform and its primary external entities: the IT Department Head, Program Coordinator, Faculty Members, MIS Staff, OJT Interns, Student Users, and the IoT Smart Key Box.

The IT Department Head interacts with the system as the super administrator, supplying curriculum structures, finalizing or reopening master timetables, managing the lifecycle of faculty and MIS staff accounts, approving or rejecting second-key reservations, and recording structured follow-up inquiries on maintenance tickets. The Program Coordinator operates as an administrative collaborator, supplying draft schedule allocations, updating faculty records, and submitting second-key requests, receiving working draft previews and conflict alerts. Faculty Members receive official timetable views and overdue reminders while submitting hallway QR key custody handoffs, advance second-key requests, and PC remarks. MIS Staff manage equipment inventories, generate workstation QR labels, and supervise OJT Interns who perform physical repairs and log diagnostic notes. Students submit verified hardware defect tickets through a dual-stage QR scan. Finally, the IoT Key Box node interfaces continuously via Wi-Fi telemetry, transmitting physical key sensor readings and barcode events while receiving real-time room states for external LCD display.

For equipment maintenance, the Student entity acts as an external reporter, submitting PC condition reports for defective desktop parts without needing an account, and receiving an immediate report submission acknowledgment. The MIS staff manages these records by providing login credentials, maintenance ticket updates, QR generation requests, and PC report search queries. In return, the system outputs authentication status, ticket status feedback, printable PC asset QR codes, and filtered PC issue results to guide physical computer repairs across the laboratories.

Data Flow Diagram



The Level 1 Data Flow Diagram (DFD) in Figure 4 explains how information moves inside LabSync through five primary operational processes: User & Access Management (1.0), Scheduling & Room Monitoring (2.0), Equipment Maintenance & Verification (3.0), Reporting & Analytics (4.0), and Key Custody & Authorization Management (5.0).

User & Access Management (1.0) handles authentication, session validation, and role-based permissions across the five user roles. Credentials are validated against the Users Store (D1). The IT Department Head manages Faculty and MIS Staff accounts - including soft deactivation and one-active-technician replacement workflows - while MIS Staff provision temporary OJT Intern accounts with defined date limits.

Scheduling & Room Monitoring (2.0) manages timetable construction and real-time laboratory status tracking. The IT Department Head and Program Coordinator collaborate on timetable construction by storing working draft blocks in the Schedule Drafts Store (D8). Draft modifications are protected by in-memory room lease locks and version tracking in Schedule Metadata (D9), preventing premature alteration of active classes. The system validates entries against Curriculum Subjects (D3) and Laboratories (D5) to detect instructor double-booking and room overlaps. Only the IT Department Head possesses the administrative authority to finalize drafts into the official Schedules Store (D2) or reopen finalized schedules for revision.

Equipment Maintenance & Verification (3.0) processes hardware tracking and repair tickets for laboratory workstations. To ensure ticket integrity without requiring student login accounts, students perform a dual QR scan: scanning the PC sticker followed by their institutional Student ID. The system generates a single-use 15-minute verification nonce in Student Verification Sessions (D10) and automatically populates the student's name and ID number before committing the ticket to the Maintenance Store (D7). MIS Staff and OJT Interns review tickets, inspect hardware histories in Lab Units (D6), and submit repair diagnostic notes. The IT Department Head can record once-per-calendar-day follow-up inquiries on unresolved tickets.

Reporting & Analytics (4.0) handles data retrieval and summary generation across the laboratories. It queries the Lab Units Store (D6) for inventory health and pulls active fault logs from the Maintenance Store (D7). From this combined data, the process delivers printable PC maintenance reports to the Department Head, filtered issue lists to MIS Staff for inspections, and room maintenance summaries to Faculty to help them monitor equipment conditions in their classrooms.

Key Custody & Authorization Management (5.0) tracks key custody and governs physical room access. Faculty members and the Program Coordinator can submit advance second-key reservation requests for authorized academic sessions up to the following Saturday, stored in Key Authorization Requests (D11) and reviewed exclusively by the IT Department Head. Physical key handoffs between consecutive instructors utilize mobile QR scans, updating key custody in Laboratories (D5). In addition, physical key insertion and withdrawal events are captured in real time via the IoT Key Box hardware node.

Use Case Diagram

■ = Implemented / Updated Feature

Figure 5. Use Case Diagram – LabSync

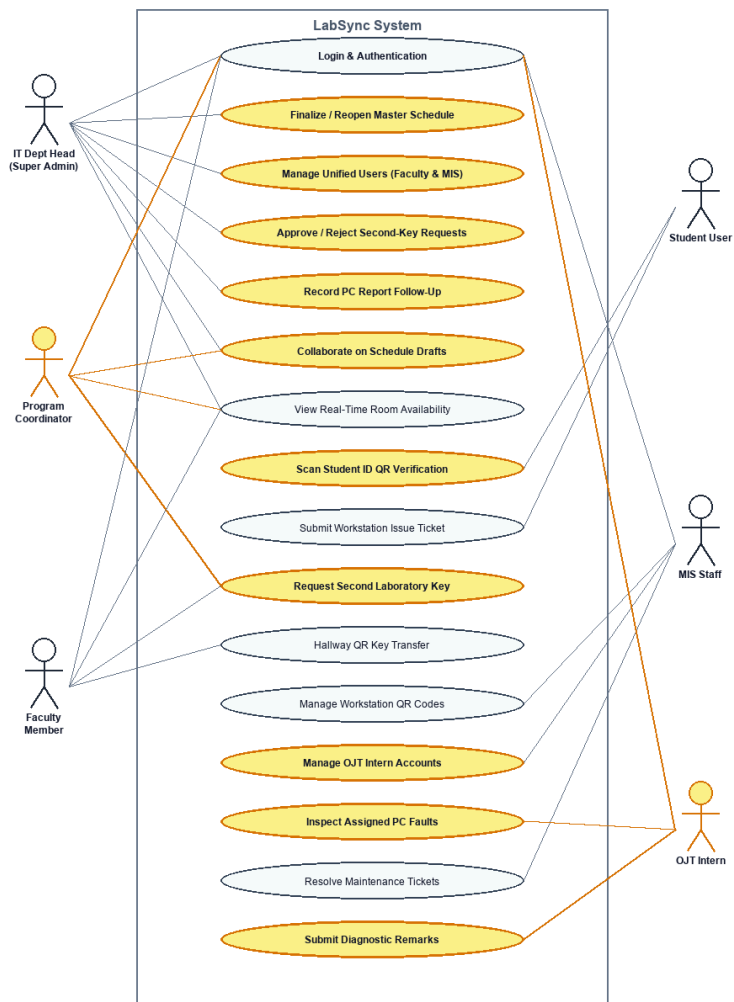


Figure 5. Use Case Diagram

The Use Case Diagram displays the system boundary and interactions between the different users of LabSync—Department Head, Faculty, MIS Staff, and Student. The system segregates its features according to the roles as that every person has access only to what he or she needs for his or her task.

The Use Case Diagram in Figure 5 delineates the functional scope and operational boundaries across all six user roles: IT Department Head, Program Coordinator, Faculty Members, MIS Staff, OJT Interns, and Students.

The IT Department Head exercises administrative authority over core institutional assets, possessing exclusive permissions to finalize or reopen master timetables, manage the unified User Management module (including faculty registration and MIS Staff deactivation/replacement), approve or reject faculty second-key requests, record once-per-calendar-day PC report follow-up inquiries, and transfer leadership.

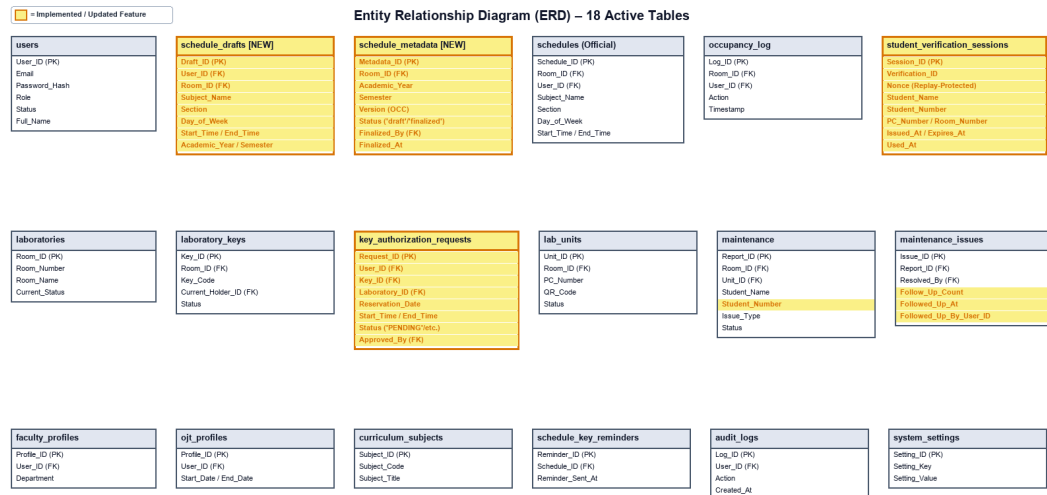
The Program Coordinator functions as an administrative collaborator with bounded authority. The coordinator collaborates on draft class timetables within isolated draft storage, monitors room availability, manages faculty profile records, and can request an additional laboratory key. However, the coordinator is strictly prevented from finalizing schedules, approving other key requests, or managing technical staff.

Faculty Members access personal class timetables, monitor room availability, execute hallway QR key custody handoffs, and submit advance second-key reservation requests for authorized sessions up to the following Saturday.

MIS Staff oversee hardware maintenance tickets, generate workstation QR labels, and provision temporary OJT intern accounts with fixed dates. OJT Interns log in to view assigned computer repair tickets and record diagnostic resolution remarks, while smart key boxes and scheduling tools remain strictly restricted.

Students utilize the zero-login workflow with dual QR verification: scanning the workstation QR code followed by their Student ID QR code to verify identity, entering their class section, and submitting defect reports.

Entity Relationship Diagram (ERD)



The Entity Relationship Diagram (ERD) in Figure 6 illustrates the relational data structure of LabSync, encompassing 18 active tables established across verified database migrations.

Core user accounts and permissions are maintained in the users table, linked to faculty_profiles and ojt_profiles. Physical laboratories and hardware assets are structured around the laboratories table, which connects to individual computer workstations in lab_units and physical key records in laboratory_keys.

Timetable management utilizes an isolated two-tier structure: collaborative working schedules are stored in schedule_drafts, while published class schedules reside in schedules. Transition between draft and official status is governed by schedule_metadata, which tracks room-level version numbers (optimistic concurrency control), draft status ('draft' or 'finalized'), and the finalizing Department Head ID. Key custody transactions are recorded in occupancy_log, while advance reservations are managed in key_authorization_requests.

Maintenance workflows center on the maintenance table, which stores workstation fault reports populated with Student_Number via 15-minute single-use sessions in student_verification_sessions. Diagnostic resolutions and IT Head follow-up inquiries (tracking Follow_Up_Count, Followed_Up_At, and user attribution) are maintained in maintenance_issues.

Tables like `audit_logs` and `iot_devices` simply record background activity and keep the ESP32 hardware synced with the app.

Circuit Diagram & Pin Mapping Table

Circuit Diagram

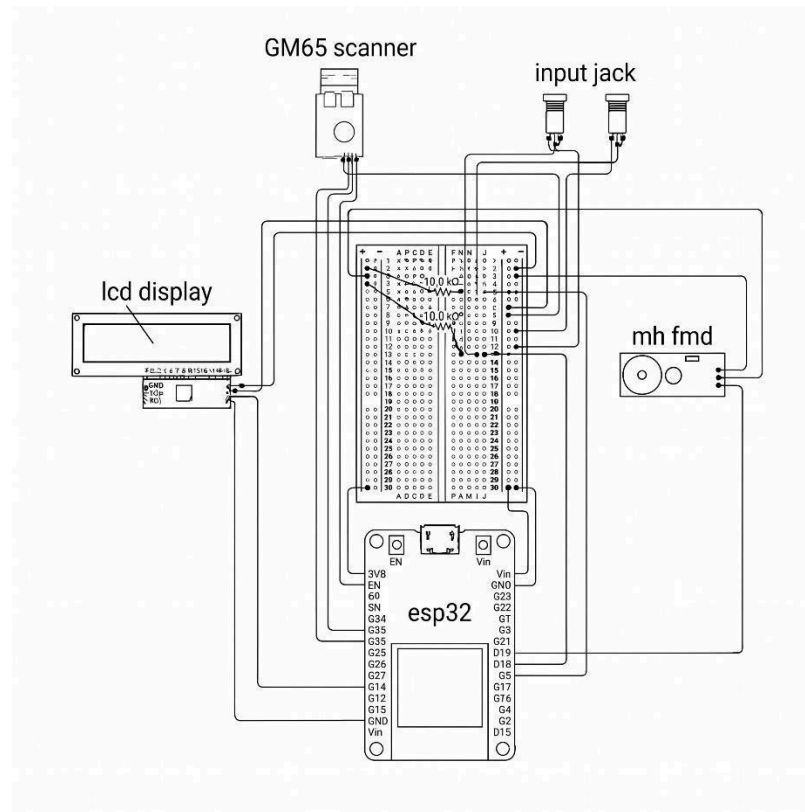


Figure 6. Circuit Diagram

The circuit diagram details the schematic wiring and signal routing of the system, governed by an ESP32 microcontroller that manages all data processing. Input data is transmitted from a GM65 Scanner via serial communication for code scanning, while two input jacks are connected across a

breadboard utilizing dual $10\text{k}\Omega$ pull-up resistors to enforce stable logic voltage levels on the input pins. For output handling, the microcontroller interfaces with an LCD Display over I2C communication (SDA/SCL lines) for visual status updates, and drives an MH-FMD Buzzer Module to output audible alert signals, with all operational modules tied to common power and ground rails.

Pictorial Diagram

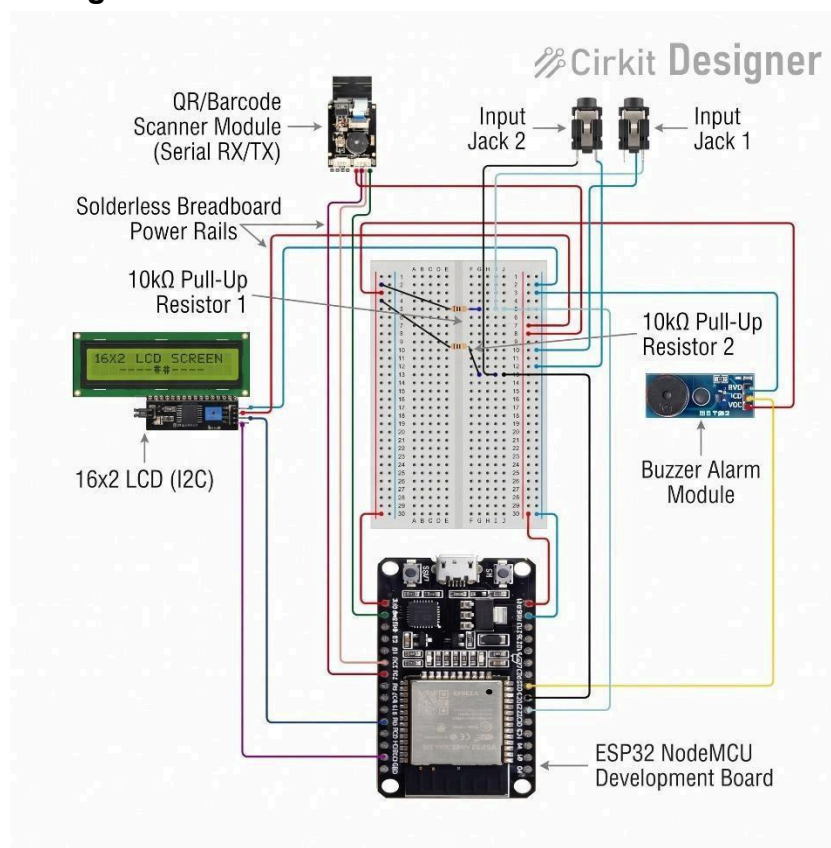


Figure 7. Pictorial Diagram

The pictorial diagram illustrates the complete physical hardware integration of the system, centered around an ESP32 NodeMCU Development Board which serves as the main processing unit. System input is captured

through a QR/Barcode Scanner Module using Serial RX/TX pins for credential reading, alongside Input Jack 1 and Input Jack 2 which interface with external sensors, stabilized by two 10k Ω pull-up resistors on a solderless breadboard to prevent floating signal states. For visual feedback and status monitoring, the microcontroller is wired to a 16x2 LCD display integrated with an I2C module to simplify data lines, while a Buzzer Alarm Module provides immediate auditory feedback during system operations. All peripheral components receive power and ground connections routed directly through the solderless breadboard power rails, establishing a unified hardware circuit.

Pin Mapping Table

Table 5. Pin Mapping Table

Component	Pin Label	ESP32 Pin (GPIO)	Function / Description
Red LED (10mm)	Anode (+)	GPIO 5	Output Signal (Alert / No Key)
	Cathode (-)	GND	Ground Connection
Resistor (220 Ω –330 Ω)	End A	GPIO 5	Series with Red LED
	End B	Red LED Anode	Current limiting

Green LED (10mm)	Anode (+)	GPIO 2	Output Signal (Normal / Key Present)
	Cathode (-)	GND	Ground Connection
Resistor (220Ω– 330Ω)	End A	GPIO 2	Series with Green LED
	End B	Green LED Anode	Current limiting
Magnetic Switch	Wire 1	GPIO 4	Sensor Signal (Input)
	Wire 2	GND	Ground Connection
Pull-up Resistor (10kΩ)	End A	GPIO 4	Connected to Sensor Signal
	End B	3V3	Pull-up to HIGH

The LabSync Pin Mapping Table lists the specific wiring connections between the external hardware parts and the ESP32 microcontroller to make sure the tracking system works correctly. The hardware setup is divided into a visual indicator subsystem 43 and a sensory input loop. For the visual feedback, a Red LED is connected to GPIO 5 to show alerts or that the key is missing, while a Green

LED is wired to GPIO 2 to show normal status or that the key is in its place. Both LEDs use resistors to protect the circuit and limit the electrical current. For the sensor input, a magnetic switch is connected to GPIO 4 to read whether the lab key is present. This sensor uses a 10k pull-up resistor connected to the 3V3 pin to keep the signal stable and prevent false readings when the key is moved.

System Architecture Diagram

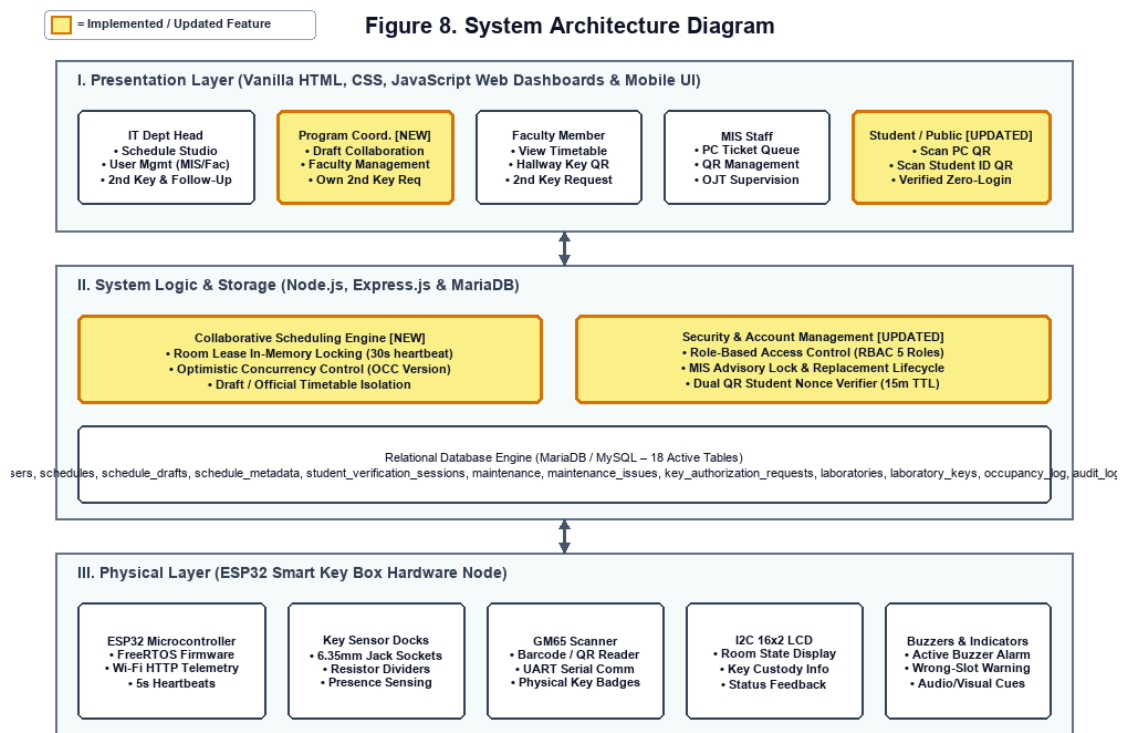


Figure 8. System Architecture Diagram

The LabSync system is built on a robust three-tier architecture connecting responsive client web interfaces, centralized application logic and storage, and physical laboratory IoT hardware nodes.

I. Presentation Layer

Developed using Vanilla HTML5, CSS3, and ES6+ JavaScript, this layer provides tailored interfaces:

- Administrative & Teaching Dashboards: Provide role-specific workspaces for the IT Department Head (Master Schedule Studio, User Management, Approvals), Program Coordinator (Collaborative Drafts, Faculty Management), Faculty Members (Schedules, Hallway QR Handoff, 2nd-Key Request), MIS Staff (Ticket Queue, QR Code Management, OJT Supervision), and OJT Interns (Assigned Repair Tickets).
- Student / Public Interface: A mobile web interface supporting friction-free defect reporting via dual QR scanning (workstation QR plus Student ID QR verification) without user registration.

II. System Logic & Storage

This centralized backend layer executes business rules, role authorization, and data persistence:

- Collaborative Draft Engine: Implements in-memory room lease locking (30-second heartbeats) and optimistic concurrency control (OCC version checks) to isolate draft edits from official schedules.
- Security & Verification Services: Governs bcrypt password hashing, 5-role RBAC authorization, MIS advisory locking for 1-active-technician guarantees, and 15-minute replay-protected student nonces.
- MariaDB Database: Houses 18 relational tables preserving schedules, maintenance histories, and logs.

- MySQL Database: Contains system's fundamental data such as user accounts, room scheduling, key transaction logs, and equipment maintenance reports submitted by students.

III. Physical Layer (Lab Node)

The physical layer includes the IoT hardware in each laboratory room used to automate important tracking and occupancy monitoring:

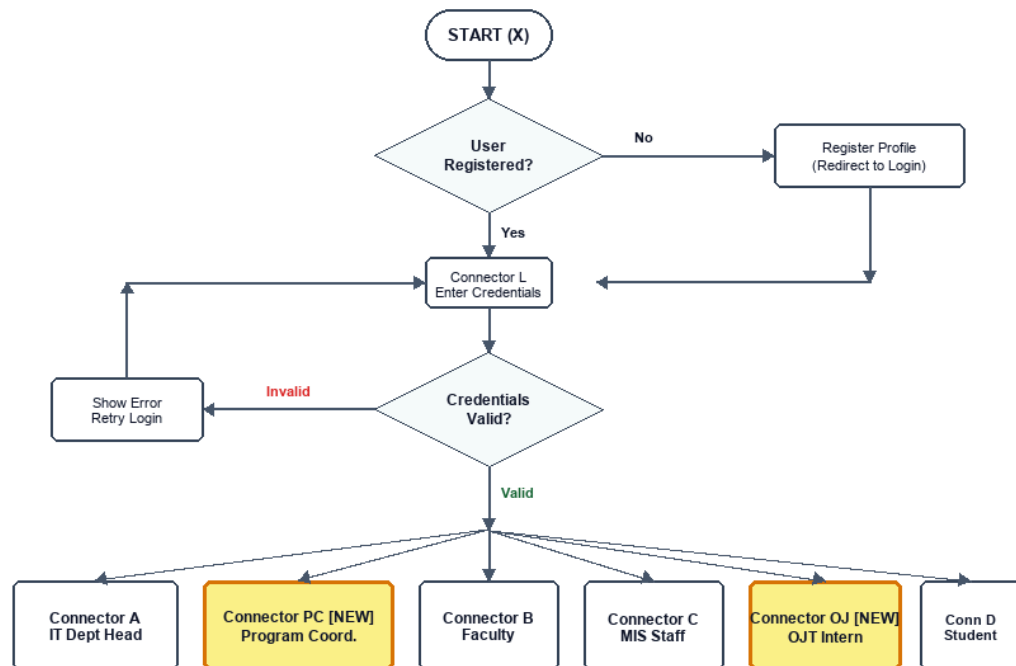
- ESP32 Microcontroller: It is to act as the brain of the node, gathering data from the sensors and providing up-to-the-minute data about the node's condition to the web server through Wi-Fi using HTTP.
- Magnetic Reed Switch: A switch that detects a physical presence of a key on the key pad, indicating if the room key is present or not.
- QR Scanner Module: Capture the QR codes and recognized from the physical world and provide the data to the Microcontroller directly.
- LCD Screen: It is used to show the room status immediately outside of the lab for instructor and student viewing.
- Active and Passive Buzzer Module: Provides audible alarms to indicate successful scan and/or unauthorized key access.

Flowchart

Login

 = Implemented / Updated Feature

Figure 9. Login Flowchart



When a user accesses LabSync, the process initiates at connector X. Registered users proceed to connector L to provide login credentials. The authentication engine verifies user credentials against stored password hashes and evaluates role permissions. If verification fails, an error notice is displayed and the user is returned to connector L. Upon successful authentication, the system directs the user to their dedicated role portal: connector A for Department Head, connector PC for Program Coordinator, connector B for Faculty, connector C for MIS Staff, or connector OJ for OJT Interns. Students accessing the workstation QR reporting workflow bypass administrative login and proceed directly to connector D.

credentials. If the credentials are incorrect, the user is sent back to connector **L** to try again. Once verified, the site redirects the user to their respective web dashboard at **A** for Department Head, **B** for Faculty, or **C** for MIS Admin. Users who enter as **Students** skip the admin login checks and go straight to connector **D**.

Department Head User

■ = Implemented / Updated Feature

Figure 10. IT Department Head Workflow Flowchart

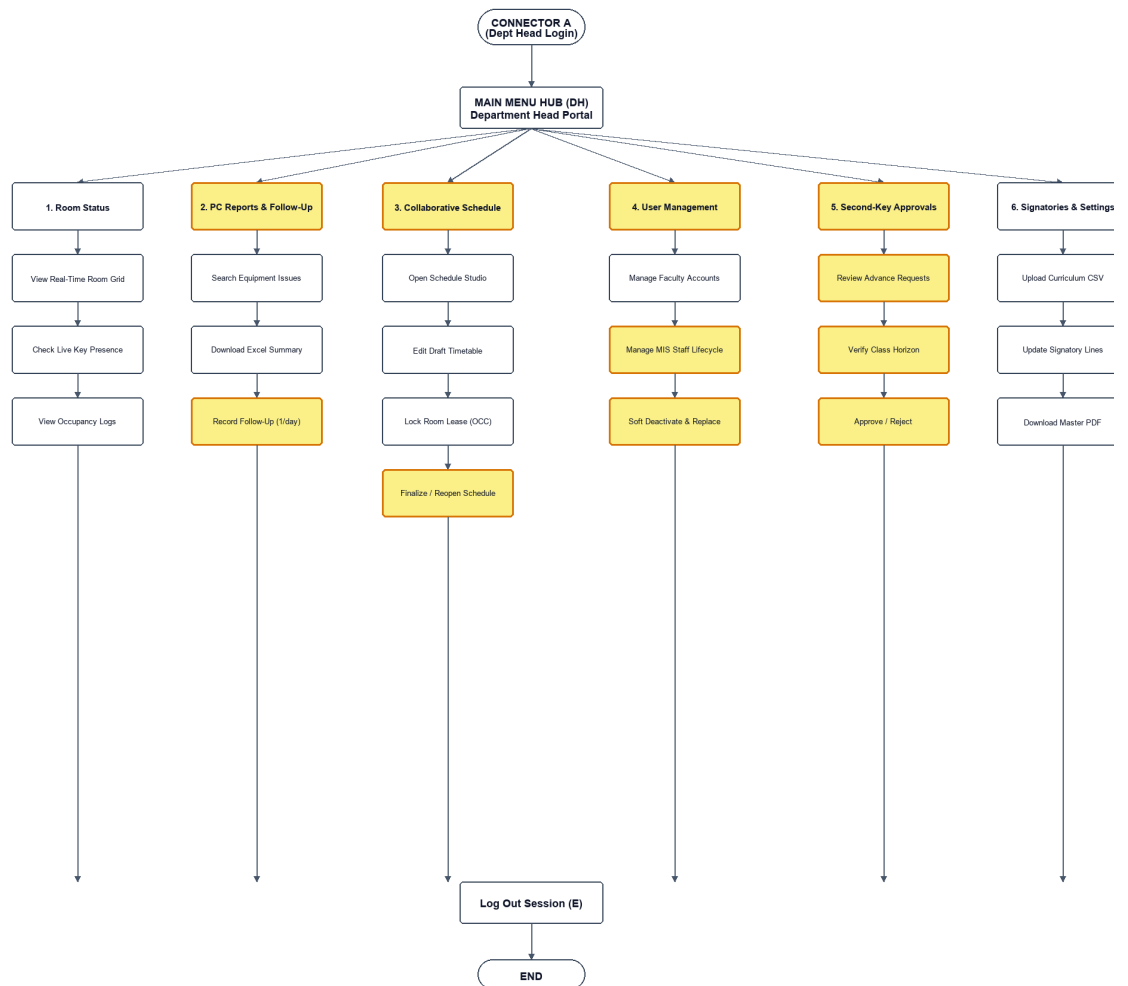


Figure 10. IT Department Head Workflow Flowchart

The Department Head workflow originates at connector A and accesses the main administrative hub (DH) to manage departmental operations:

- Room Status & PC Reports: Inspects real-time room occupancy and key presence, downloads Excel reports, and records once-per-calendar-day follow-up inquiries on unresolved maintenance tickets.
- Collaborative Master Schedule: Opens Schedule Studio to edit timetable blocks with professor conflict checks. Edits are isolated in working drafts with room lease locking. The Department Head retains sole authority to finalize drafts into the official schedule or reopen finalized schedules for adjustment.
- User Management: Consolidates administrative control over Faculty accounts and MIS Staff accounts, including soft deactivation, historical record preservation, and replacement workflows enforcing one active technician.
- Second-Key Approvals: Reviews advance laboratory key requests submitted by Faculty and Coordinators, validating class reservation dates up to the following Saturday before granting approval or rejection.
- Settings & Signatories: Manages curriculum subject codes, institutional signatory titles, and exports official timetables as signed PDFs. Logging out terminates the session at connector E.

Faculty User

■ = Implemented / Updated Feature

Figure 11. Faculty Member Workflow Flowchart

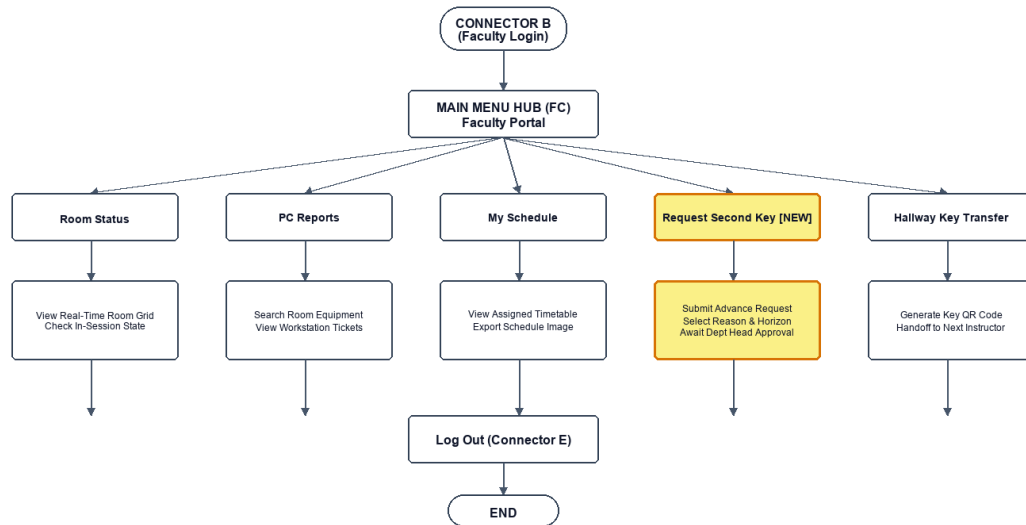


Figure 11. Faculty Member Workflow Flowchart

The Faculty workflow begins at connector B and utilizes the faculty menu hub (FC) to access instructional and laboratory tools:

- Room Status: Monitors real-time laboratory room availability (Available, In Session, Borrowed) and active keyholder identity.
- PC Reports: Searches equipment fault reports in assigned teaching rooms and downloads diagnostic logs.
- My Schedule: Reviews personalized teaching timetables filtered by academic year and semester, with visual timetable export capabilities.
- Request Second Key: Enables instructors to submit an advance reservation request for an additional laboratory key for authorized academic sessions up to the following week's Saturday, which is routed to the Department Head for review and approval.
- Hallway Key Transfer: Facilitates peer-to-peer key custody handoffs between consecutive instructors using mobile QR codes, automatically updating database records and resetting overdue reminder timers.

Clicking Log Out concludes the session and routes to connector E.

Management Information System (MIS) User

■ = Implemented / Updated Feature

Figure 12. MIS Staff Workflow Flowchart

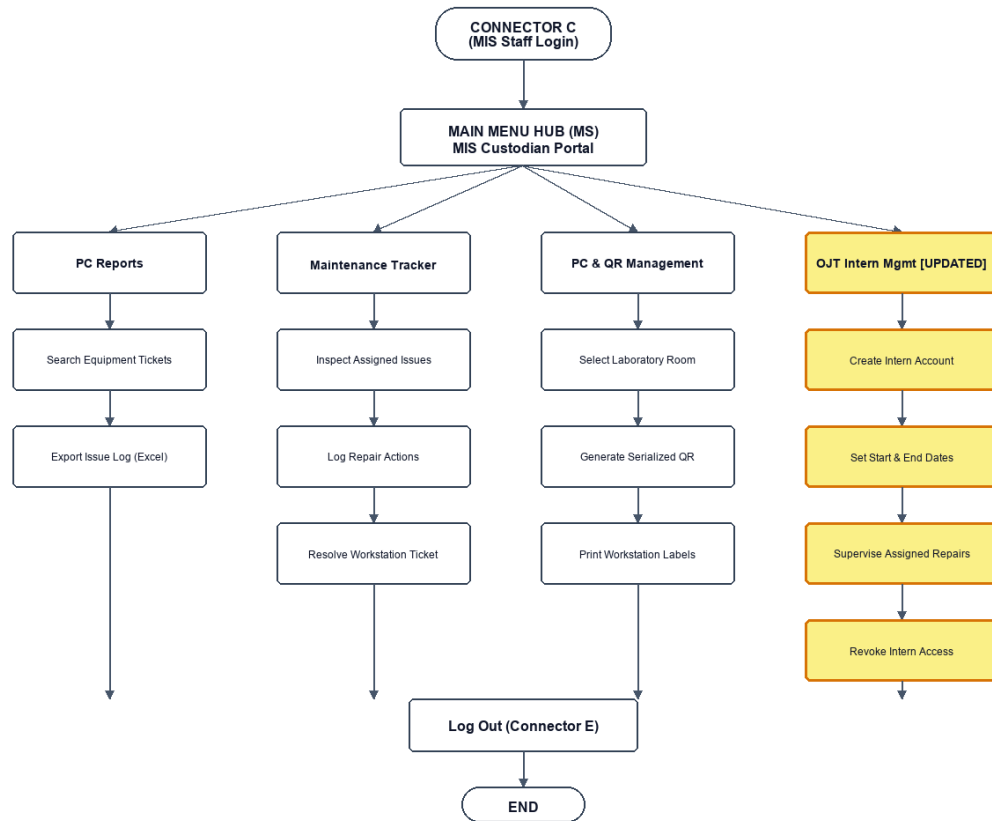


Figure 12. Management Information System (MIS) Workflow Flowchart

The MIS Staff workflow initiates at connector C and operates through the technician hub (MS) to conduct daily maintenance and hardware administration:

- PC Reports: Evaluates incoming hardware fault tickets, filters issues by component category, and exports reports to spreadsheet formats.
- Maintenance Tracker: Reviews active workstation repair tickets, logs diagnostic findings and component replacements, and resolves completed tickets.
- PC & QR Management: Selects computer laboratories to generate, download, and print serialized QR code labels for individual workstations.
- OJT Intern Management: Creates and administers temporary accounts for student interns, setting explicit start and end internship dates. Interns assist with physical repairs while administrative controls and smart key boxes remain strictly restricted. MIS accounts are provisioned exclusively by the Department Head.

Clicking Log Out ends the technician session and returns to connector E.

which creates temporary login passwords for them. Finally, clicking **Log Out** safely ends their session and returns to connector **E**.

Student User



Figure 13. Student

Figure 13. Student Zero-Login Verification & Reporting Flowchart

The zero-login reporting feature allows students to report defective computer equipment quickly while maintaining institutional accountability through dual QR verification:

1. The student scans the serialized QR code sticker attached to the laboratory workstation.
2. The interface prompts the student to scan their physical Student ID QR code.
3. The system validates the verification nonce, automatically extracting the Student Name and Student Number.
4. The student selects their class section and toggles defective components (monitor, keyboard, mouse, unit).
5. Submitting the ticket commits the record into the maintenance database with authenticated student attribution.

Upon submission, a confirmation receipt is presented and the workflow resets to connector X.

away without any hassle.

Table 6. Tools, Technologies, and Development Environment

Category	Tool/ Technology	Version
Frontend	HTML5 CSS3 Vanilla JavaScript	HTML5 CSS3 ES6+
Backend	<u>Node.js</u> Express.js	Node.js v18.0.0+ Express 5

Database	MySQL MariaDB	8.4.8 (LTS)
IDE/Code Editor	Visual Studio Code	1.112
Diagram Tools	Draw.io	Web-based (Always Latest)
UI/UX Design	CSS3 Custom Design System	3.0
Deployment	Railway	v5.49.1
Email Service	Nodemailer	V9.0.6
QR Code Engine	Node.js qrcode library	v1.5.4
Security&Auth	Bcrypt, express-session, crypto	Bcrypt v6.0.0 (12 salt rounds)
IoT Firmware	C++ (Arduino Framework / ESP32 Core)	ESP32 Core v2.0+

The Tools, Technologies, and Development Environment section outlines the software technologies, hardware frameworks, and development tools used to develop, test, and manage the LabSync system, supporting the process from initial design to system implementation. The system uses HTML5, CSS3, and Vanilla JavaScript for the frontend to provide a responsive and accessible user interface, while Node.js and Express.js handle the backend logic and API communication. MySQL MariaDB is used as the database for storing laboratory information, schedules, maintenance records, user accounts, and key management data. Draw.io is used for system diagrams and architecture planning, while the Custom Design System is used to maintain the overall interface and visual design. Furthermore, the system integrates specific libraries such as Nodemailer for email services, the Node.js qrcode library for generating QR codes, and Bcrypt alongside express-session for secure user authentication.

Finally, the IoT key box firmware is developed in C++ using the ESP32 Arduino Core, and the entire web application is deployed and managed for actual use via the Railway platform.

Development Cost

Cost and Benefit Analysis

Table 7. Human Resource Cost

DEVELOPMENT COST			
A. Human Resource Cost			
Development Team	Monthly Salary	Duration Months	Subtotal
Front-end Developer	₱25,000	5 months	₱125,000
Back-end Developer	₱28,000	5 months	₱140,000
IoT Developer	₱25,000	5 months	₱125,000
IoT Developer	₱25,000	5 months	₱125,000
QA Tester	₱22,000	5 months	₱110,000
Total:			₱625,000

The table presents the estimated human resource cost for the development of LabSync. It shows the different development team roles, their estimated monthly salary, the five-month development period, and the corresponding subtotal. The team consists of a Front-end Developer, Back-end Developer, two IoT Developers, and a QA Tester. The total estimated human resource cost for the five-month development period is ₱625,000.

Table 8. Software & Licensing Cost

DEVELOPMENT COST				
B. Software and Licensing	UNIT COST	QUANTIT Y	MONTH S	SUBTOTAL
ITEM NAME				
Cloud Web Hosting	₱3,763.20	1	12	₱3,763.20
Domain Name Registration	₱449.28	1	12	₱449.28
Visual Studio Code	₱0.00	1	5	₱0.00
Node.js	₱0.00	1	5	₱0.00
Arduino IDE	₱0.00	1	5	₱0.00
Figma	₱0.00	1	5	₱0.00
XAMPP	₱0.00	1	5	₱0.00
Draw.io	₱0.00	1	5	₱0.00
TOTAL:	₱4,212.48			

The table presents the estimated software and licensing costs required for the development and deployment of LabSync. It includes the cost of cloud web hosting through Railway and domain name registration, which are necessary for making the system accessible online. The development team also uses Visual Studio Code, Node.js, Arduino IDE, Figma, XAMPP, and Draw.io during the five-month development period. These tools are listed at ₱0.00 because the versions used by the team are available at no licensing cost. The total estimated software and licensing cost is ₱4,212.48.

Table 9. Hardware Cost

DEVELOPMENT COST			
C. Hardware Cost	UNIT COST	QUANTIT Y	SUBTOTAL
ITEM NAME			
GM65 QR Code Recognition Module	₱3,763.20	1	₱1,055.00
ESP32 Dev Module	₱198.00	1	₱198.00
16x2 Character LCD Module with I2C	₱119.00	1	₱119.00
Active and Passive Buzzer Module	₱32.00	1	₱32.00
Wall Mounted key plugs 4 pcs	₱114.00	1	₱114.00
40-Pin Jumper Wire (Male - Male, 20 cm)	₱39.00	2	₱78.00
40-Pin Jumper Wire (Male - Female, 20cm)	₱68.00	1	₱68.00
6.35 mm ¼ Guitar Input Jack Copper Mono, Female Connector	₱47.00	8	₱376.00
Resistor Through-Hole 1/4w 10pcs	₱20.00	1	₱20.00
16x2 LCD Bezel/Panel Mounting Frame	₱109.00	1	₱109.00
Heat Shrink Tube Assortment wrap wire	₱28.00	10	₱28.00
Blank Acrylic key chain 4.5*2.8cm 10 pcs	₱59.00	1	₱59.00
Acrylic Case and Manual Labor fee	₱1,700.00	1	₱1,700.00
Sticker Design Acrylic Case	₱770.00	1	₱770.00
Total:			₱4,726.00

The hardware cost covers the electronic components, wiring materials, mounting parts, and fabrication materials required to develop and assemble the LabSync laboratory monitoring system. These components support the system's QR code recognition, key availability monitoring, information display, and notification functions. The ESP32 Dev Module serves as the main microcontroller, while the GM65 QR Code Recognition Module is used to scan QR codes. The LCD module provides system information and status messages, while the buzzer provides audible notifications. Jumper wires, resistors, heat shrink tubes, and connectors are used for electrical connections and circuit assembly. The wall-mounted key plugs, acrylic key chains, acrylic case, and sticker design are

used for the physical key management and enclosure of the device. The total estimated hardware development cost is **₱4,726.00**.

Table 10. Utilities & Resource Cost

DEVELOPMENT COST				
A. Utilities & Resource Cost				
Development Team	Unit Cost	Quantity	Months	SUBTOTAL
Laptop	₱25,000.00	5	5	₱100,000
Transportation	₱500.00	1	5	₱2,500.00
Internet	₱1,500.00	1	5	₱7,500.00
Electricity	₱1,000.00	1	5	₱5,000.00
Printer & Supplies	₱6,000.00	1	1	₱6,000.00
Total:				₱121,000.

The Utilities and Resource Cost covers the resources and basic expenses required during the development, testing, and documentation of the LabSync system. Laptop or computer usage is allocated for the five development team members to support programming, database management, system testing, and documentation. Transportation covers project-related travel such as laboratory visits, meetings, and system testing. Internet and electricity are necessary for development, communication, database access, and operation of the IoT devices. Printer and supplies are included for printing questionnaires, documentation, testing materials, and other project requirements. The estimated total Utilities and Resource Cost is **₱121,000**.

Table 11. Summary of Total Development Cost

Summary of Total Development Cost	
Particulars	Cost
A. Human Resource	₱625,000.00
B. Software & Licensing	₱4,212.48
C. Hardware Cost	₱4,726.00
D. Utilities & Resource Cost	₱121,000.00
Total:	₱754,938.48

The total one-time development cost represents the estimated overall expense required to develop, test, and implement the LabSync system. It includes the costs for the development team, software and licensing, hardware components, and utilities and resources used throughout the project. Based on the estimated expenses under each category, the total one-time development cost of LabSync is **₱754,938.48**.

Evaluation Methods

This section describes how the system will be evaluated following implementation. The objective of the evaluation is to assess the technical quality of the system and its acceptability to users.

To achieve this, two established evaluation models will be used:

- **ISO/IEC 25010 Software Product Quality Model**

- **Technology Acceptance Model (TAM)**

Evaluation Using ISO/IEC 25010

The ISO/IEC 25010 Software Quality Model was used to evaluate the technical quality of the LabSync. This evaluation will be conducted by Information Technology (IT) Experts who have knowledge and experience in system development and software evaluation.

Purpose: The evaluation aims to assess the software quality of LabSync based on its functional suitability, performance efficiency, usability, and reliability.

Table 12. Evaluation Using ISO/IEC 25010

Criteria	Description
Functional Suitability:	measures the extent to which LabSync provides the required functions for laboratory availability monitoring, scheduling, PC issue reporting, maintenance management, key monitoring, and other system operations.
Usability:	measures how easy the system is to understand, learn, navigate, and operate for its intended users.
Reliability:	evaluates the system's ability to perform its required functions consistently and maintain stable operation during normal use.
Performance Efficiency:	to measure the system's responsiveness in displaying real time data.
Maintainability:	assesses how easy it is to modify, update, and improve the system when necessary.

Portability:	evaluates the system's ability to operate in different environments, devices, or platforms with minimal adjustments.
Compatibility:	Evaluates the system's ability to operate effectively alongside other systems, software, and devices without causing conflicts or affecting their normal operation.
Security:	Evaluates the system's ability to protect user information, system data, accounts, and functions from unauthorized access and other security threats.
Respondents:	a minimum of 3 IT Professionals will be selected to evaluate the system.
Instrument:	an evaluation questionnaire will be prepared based on the ISO/IEC 25010 software quality characteristics. A 5-point Likert Scale (1 = Strongly Disagree to 5 = Strongly Agree) will be used for each item.

Evaluation Using Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) will be used to evaluate the system from the perspective of the actual end-users, specifically the students, faculty members, and MIS personnel of Bulacan State University – Sarmiento Campus.

Purpose: The evaluation aims to determine the users' perceptions of the usefulness, ease of use, behavioral intention, and actual system use of LabSync in supporting laboratory scheduling, room availability monitoring, PC issue reporting, maintenance management, and laboratory key monitoring.

The TAM evaluation will consider the following areas:

Perceived Usefulness (PU): measures the extent to which users believe that LabSync helps them perform laboratory-related tasks more effectively and efficiently.

Perceived Ease of Use (PEOU): evaluates how easy users find LabSync to learn, understand, navigate, and operate.

Behavioral Intention to Use (BI) – measures the users' willingness or intention to continue using LabSync for laboratory-related activities.

Actual System Use (AU) – measures the extent to which users actually use LabSync and its available functions during their laboratory-related activities.

Respondents: Fifteen (15) end-users consisting of faculty members, MIS staff, and students who have interacted with the LabSync system will participate in the evaluation.

Sampling Method: Purposive sampling will be used to ensure that respondents have prior experience using the system.

Instrument: A TAM-based questionnaire with a 5-point Likert scale (1 – Strongly Disagree to 5 – Strongly Agree) will be used to collect responses.

Data Handling and Analysis

After collecting the evaluation responses, the data will be organized and analyzed to determine the overall acceptance of the LabSync system. The following procedures will be applied:

Tabulation: All responses from the evaluation forms will be compiled and summarized into tables.

Mean Computation: The mean score for each evaluation criterion and TAM construct will be calculated.

Mean Score Range	Interpretation	Performance Evaluation
4.51 – 5.00	Highly Acceptable	The system strongly meets the expected user and operational requirements.
3.51 – 4.50	Acceptable	The system meets the expected user and operational requirements.
2.51 – 3.50	Moderately Acceptable	The system meets basic requirements but may require some improvement.
1.51 – 2.50	Slightly Acceptable	The system only partially meets the expected requirements and requires improvement.

1.00 – 1.50	Not Acceptable	The system does not sufficiently meet the expected requirements.
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